

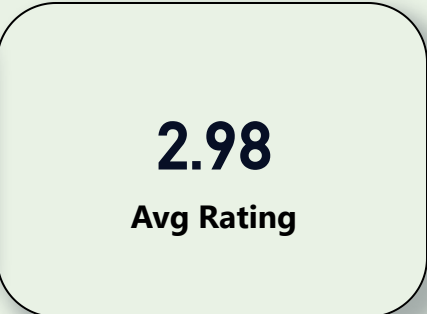
Executive Overview Dashboard

price_category

Budget (<£20)

Mid-Range (£20-£40)

Premium (>£40)



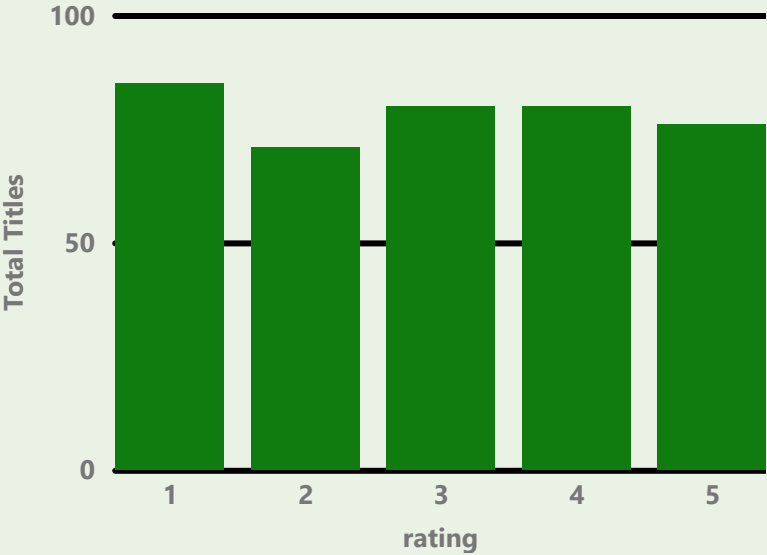
Total Titles by price_category

Premium (>£40)

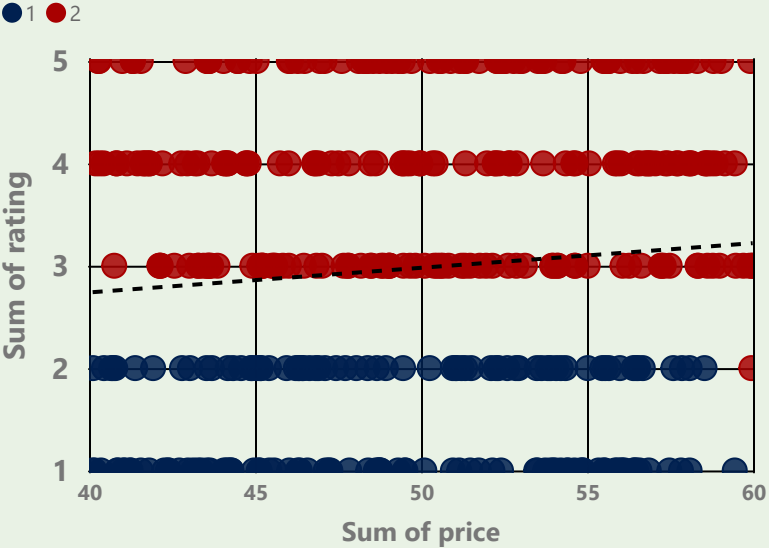


392

Total Titles by rating



Sum of price and Sum of rating by title and inventory_cluster



Cluster Diagnostics Dashboard



price_category

Budget (<£20)

Mid-Range (£20-£40)

Premium (>£40)

title	Sum of price	Sum of rating	Sum of stock_units	Sum of inventory_value
"Most Blessed of the Patriarchs": Thomas Jefferson and the Empire of the Imagination	44	5	1	44
#GIRLBOSS	51	1	1	51
10-Day Green Smoothie Cleanse: Lose Up to 15 Pounds in 10 Days!	50	5	1	50
11/22/63	48	3	1	48
1st to Die (Women's Murder Club #1)	54	1	1	54
32 Yolks	54	2	1	54
A Brush of Wings (Angels Walking #3)	56	1	1	56
A Game of Thrones (A Song of Ice and Fire #1)	46	2	1	46
A Gathering of Shadows (Shades of Magic #2)	45	4	1	45
A Heartbreaking Work of Staggering Genius	54	5	1	54
A Hero's Curse (The Unseen Chronicles #1)	50	3	1	50
Total	19555	1167	392	19555

Insights

- **Data Acquisition (Task 1):** Scraped 100% of the target data across 50 pages from books.toscrape.com, collecting exactly 1,000 clean, unblocked records.
- **Feature Engineering (Task 2):** Standardized prices into numeric floats and created commercial tiers; catalog share is dominated by Mid-Range (~40%) and Premium (~39%) books.
- **Exploratory Data Analysis (Task 3):** Customer ratings are uniformly distributed from 1 to 5 stars, showing a negligible correlation (0.03) between book price and buyer rating.
- **Machine Learning & Segmentation (Task 4):** Random Forest yielded a baseline MAE of £12.08; K-Means segmented inventory into three actionable tiers: Budget High-Turnover, Mid-Tier Low-Rating Risk, and High-Margin Premium.
- **Business Intelligence (Task 5):** The majority of the ~£34K total inventory value is concentrated in premium titles, tracked via Sync Slicers, RLS, and Drill-through diagnostics.
- **Commercial Strategy (Task 6):** Apply 15% promotional markdowns on low-rated mid-tier titles to free up locked capital, and bundle high-rated premium books with budget titles to raise average order value.